The True Cost of Volatility Targeting: Decomposing the Insurance Premium into Opportunity and Transaction Components*

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Abstract

Volatility targeting (VT) strategies scale equity exposure inversely to expected volatility, producing well-documented reductions in tail risk but also persistent return shortfalls relative to buy-and-hold. We propose a decomposition of this return gap—the "insurance premium"—into two economically distinct components: *opportunity cost* (foregone returns from reduced equity participation) and *direct cost* (transaction expenses from weight rebalancing). Using daily S&P 500 data conditioned on the CBOE VVIX index over the VVIX-reliable period 2012–2024 (N = 3,262 trading days), we find that opportunity cost accounts for 91% of the total insurance premium under unconditional VT (4.20% vs. 0.43% per annum). VVIX-conditional targeting—activating VT only when the volatility-of-volatility z-score exceeds 1.0—reduces total cost by 74% in the full sample (from 4.62% to 1.22%), primarily by eliminating unnecessary hedging during calm markets. Cross-OOS analysis shows that this cost reduction is sample-dependent (S2 outperforms in 1 of 4 non-overlapping windows), suggesting VoV conditioning operates as infrequent tail insurance rather than a consistently superior strategy. We further show

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that the 50/50 SPY/GLD benchmark generates a structural rebalancing premium of 54 basis points per annum, creating a high bar for VT to clear. While VoV-conditional VT (S2) achieves a higher Sharpe ratio than 50/50 (0.63 vs. 0.50), the comparison is not apples-to-apples: S2 is 100% equity with occasional VT, while 50/50 includes gold diversification. These findings redirect the VT design problem from minimizing transaction costs to minimizing opportunity costs through regime conditioning.

Keywords: volatility targeting, insurance premium, opportunity cost, transaction cost, volatility-of-volatility, rebalancing premium

JEL Classification: G11, G12, G17

1 Introduction

Volatility targeting (VT) has become a standard technique for dynamic risk management. The framework, formalized by Moreira and Muir (2017), prescribes scaling equity exposure inversely to conditional volatility—increasing allocation when markets are calm and reducing it when turbulence rises. Harvey et al. (2018) demonstrate that such strategies meaningfully reduce tail risk, and several studies confirm VT's ability to compress the return distribution, particularly in the left tail (Fleming et al., 2001; Hocquard et al., 2013).

Yet VT strategies systematically underperform buy-and-hold benchmarks in terms of cumulative returns. This shortfall is commonly attributed to trading friction: Harvey et al. (2018) prominently report turnover metrics as a key performance drag, and Cederburg et al. (2020) document poor out-of-sample performance across 103 volatility-managed strategies. However, this attribution conflates two fundamentally different economic mechanisms. The first is the *opportunity cost* of holding cash instead of equities during periods of elevated volatility that do not materialize into realized losses. The second is the *direct cost* of portfolio turnover—brokerage fees, bid-ask spreads, and market impact. Despite a substantial literature on VT mechanics (Moreira and Muir, 2017; Barroso and Santa-Clara, 2015; Cederburg et al., 2020; Bongaerts et al., 2020; Liu et al., 2019), the decomposition of this insurance premium into its constituent components has received little empirical attention.

While Bongaerts et al. (2020) propose conditional volatility targeting by conditioning on volatility states to optimize Sharpe ratios, our approach conditions on volatility-of-volatility and focuses on cost decomposition rather than performance optimization. This distinction matters for at least three reasons. First, our decomposition clarifies the economic nature of VT: investors are paying primarily for foregone upside, not for trading friction. Second, it implies that cost reduction should target the opportunity component rather than the transaction component, fundamentally reorienting strategy design. Third, it explains a persistent empirical puzzle: why does VT fail to outperform simple 50/50 SPY/GLD diversification, even when VT produces superior risk metrics?

We answer this question with two contributions. First, we apply a simple accounting decomposition of the VT insurance premium into opportunity and direct cost components (Section 2). While the decomposition is arithmetically straightforward, its empirical magnitude—91% opportunity cost at conservative friction assumptions—has not been previously documented and challenges the prevailing focus on transaction cost minimization. Second, we show that conditioning on the CBOE VVIX—a real-time measure of volatility-of-volatility—reduces the total premium by 74%, primarily through the opportunity cost channel (Section 3). As supporting evidence, we quantify the structural rebalancing premium of 54 basis points embedded in the 50/50 benchmark, highlighting the high bar that VT must clear to outperform simple diversification.

2 Methodology and Data

2.1 Insurance Premium Decomposition

Let r_t^{BH} denote the daily return of the buy-and-hold equity benchmark and r_t^{VT} the daily return of the volatility-targeted strategy. The VT strategy allocates weight $w_t = \min(\sigma^*/\hat{\sigma}_t, 1)$ to equities, where σ^* is the target volatility and $\hat{\sigma}_t$ is the conditional volatility estimate available at time $t-1$. We use the 12/VIX rule (Perchet et al., 2015): $w_t = \min(12/\mathrm{VIX}_{t-1}, 1)$.

Define the *total insurance premium* as the annualized return difference, where $\bar{r}$ denotes the annualized mean daily return:

$$\mathrm{IP}_{\mathrm{total}} = \bar{r}^{\mathrm{BH}} - \bar{r}^{\mathrm{VT}}. \tag{1}$$

We apply a simple accounting decomposition into two components. Let $r_t^{\mathrm{VT,gross}}$ denote the VT return *before* transaction costs are deducted. The *opportunity cost* captures the return gap due to reduced equity participation:

$$\mathrm{IP}_{\mathrm{opp}} = \bar{r}^{\mathrm{BH}} - \bar{r}^{\mathrm{VT,gross}}. \tag{2}$$

The *direct cost* captures the cumulative transaction expense:

$$\mathrm{IP}_{\mathrm{direct}} = \sum_{t} c \cdot |w_t - w_{t-1}|/N, \tag{3}$$

where $c = 5$ basis points per unit weight change, annualized over N years. By construction, $\mathrm{IP}_{\mathrm{total}} = \mathrm{IP}_{\mathrm{opp}} + \mathrm{IP}_{\mathrm{direct}}$.

2.2 VoV-Conditional Targeting

Rather than applying VT unconditionally, we propose conditioning on the volatility-of-volatility (VoV) regime. Using the CBOE VVIX index (CBOE, 2014), we compute an expanding-window z-score (minimum 60 observations):

$$z_t^{\mathrm{VoV}} = \frac{\mathrm{VVIX}_t - \overline{\mathrm{VVIX}}_{1:t}}{\mathrm{sd}(\mathrm{VVIX}_{1:t})}. \tag{4}$$

We define two conditional strategies:

- **S2 (VoV-Conditional)**: Apply 12/VIX weight only when $z_t^{\mathrm{VoV}} > 1.0$ *and* VIX is rising (5-day change > 0); otherwise hold 100% equity. The 5-day window balances responsiveness against noise: shorter windows (1–2 days) produce excessive whipsaw, while longer windows (10+ days) delay regime detection. Sensitivity to the threshold z-score is reported in Section 3.

- **S3 (Smooth VoV)**: Apply 12/VIX weight at all times, but blend toward 100% equity when $z_t^{\mathrm{VoV}} < 1.0$ via $w_t = w_t^{\mathrm{VT}} + (1 - w_t^{\mathrm{VT}}) \cdot \mathbb{1}(z_t^{\mathrm{VoV}} < 1.0) \cdot 0.5$. The 0.5 blending coefficient represents the midpoint between full VT and full equity; alternative values (0.3, 0.7) yield qualitatively similar cost decompositions.

The economic intuition is that high VoV signals regime uncertainty—periods when insurance is most valuable—while low VoV indicates stable volatility regimes where VT hedging is unnecessary and costly.[1]

[1]Unreported results show that the VoV z-score threshold alone accounts for approximately 80% of the cost reduction, with the VIX-rising filter contributing the remaining 20% by preventing premature

2.3 Data

We use daily closing prices for SPY, GLD, VIX, and VVIX from Yahoo Finance (via `yfinance`), consistent with CRSP to within rounding precision, covering the VVIX-reliable period 2012-01-03 to 2024-12-31 (N = 3,262 trading days, 12.94 years). The VVIX restriction to post-2012 data follows from documented liquidity concerns in the VVIX options market prior to this date. All signals are lagged one trading day (`signal.shift(1)`) to prevent look-ahead bias. Following a conservative assumption, transaction costs are set at 5 bps per leg on each weight change, which exceeds SPY's typical bid-ask spread of 1–2 bps (Hasbrouck, 2009) and thereby provides an upper bound on the direct cost component. At 1 bps, the opportunity cost share rises to 97%, further strengthening our core finding. We designate the out-of-sample (OOS) window as 2023-01-01 to 2024-12-31.

3 Results

3.1 Strategy Performance

Table 1 reports performance metrics for five strategies over the full sample (2012–2024). The buy-and-hold benchmark (S0) achieves a CAGR of 12.51% with a Sharpe ratio of 0.60. Unconditional VT (S1, Always 12/VIX) sacrifices 5.40 percentage points of CAGR for a 55% reduction in maximum drawdown (MDD: −15.46% vs. −34.10%). The VoV-conditional strategy (S2) retains most of the BH return (CAGR 11.14%) while still reducing MDD by 33% and achieving the highest Sharpe ratio (0.63) among all strategies.

activation during declining-volatility episodes. We acknowledge that the 80/20 attribution has not been validated through nested out-of-sample testing. It is presented as a directional estimate rather than a precise decomposition.

Table 1: Strategy Performance: Full Sample (2012–2024)

Metric	S0: BH SPY	S1: Always 12/VIX	S2: VoV-Cond.	S3: Smooth VoV	S4: 50/50 SPY/GLD
CAGR (%)	12.51	7.11	11.14	8.84	7.89
Ann. Vol (%)	16.56	9.33	13.72	11.57	11.47
Sharpe	0.60	0.54	0.63	0.57	0.50
MDD (%)	−34.10	−15.46	−22.78	−23.20	−21.17
Calmar	0.37	0.46	0.49	0.38	0.37
Sortino	0.55	0.49	0.61	0.53	0.46
Mean CRRA $\gamma = 5$	0.217	0.190	0.211	0.208	0.157
Avg. Weight (%)	100.0	74.1	93.4	86.0	—
Ann. Turnover (%)	0.0	872.4	1044.7	912.4	—

Notes: S0 is 100% SPY buy-and-hold. S1 sets $w = \min(12/\mathrm{VIX}_{t-1}, 1)$ daily. S2 applies S1 only when VoV z-score > 1.0 and VIX is rising; otherwise $w = 1$. S3 blends S1 toward full equity when VoV z-score < 1.0. S4 is 50/50 SPY/GLD with monthly rebalancing. All signals lagged one day. TX cost: 5 bps per weight change. CRRA $\gamma = 5$ reports mean CRRA utility of the wealth path (not certainty-equivalent); a formal CE analysis requires computing $\mathrm{CE} = u^{-1}(E[u(W)])$. Data source: yfinance; experiment K811v2.

3.2 Insurance Premium Decomposition

Table 2 presents the central result: the decomposition of the insurance premium into opportunity and direct cost components.

Table 2: Insurance Premium Decomposition (%/year)

Strategy	Opportunity Cost	Direct Cost	Total Premium	Opp. Share (%)	Δ vs. S1 (%)
S1: Always 12/VIX	4.20	0.43	4.62	90.7	—
S2: VoV-Conditional	0.70	0.52	1.22	57.1	−73.6
S3: Smooth VoV	2.85	0.46	3.31	86.2	−28.4

Notes: Opportunity cost = $\bar{r}^{\text{BH}} - \bar{r}^{\text{VT,gross}}$ (return gap before TX). Direct cost = $\sum c|w_t - w_{t-1}|$ annualized (TX friction). Total = Opp. + Direct. Δ vs. S1 shows percentage reduction in total premium relative to unconditional VT. Data: K811v2 (2012–2024, $N = 3{,}262$).

Three findings emerge. First, for unconditional VT (S1), opportunity cost dominates at 91% of the total premium (4.20% vs. 0.43%). This overturns the common practitioner intuition that VT is expensive because of trading costs. The actual cost is foregone equity participation during elevated-VIX periods that are followed by positive returns.

Second, VoV conditioning (S2) reduces the total premium by 74%, from 4.62% to 1.22% per annum. Critically, this reduction operates almost entirely through the opportunity cost channel: S2's opportunity cost falls from 4.20% to 0.70% (an 83% reduction), while its direct cost actually *increases* slightly from 0.43% to 0.52%. The mechanism is intuitive: S2 is fully invested 86% of the time ($w = 1$), engaging VT only during the 14% of days when VoV z-score exceeds 1.0 and VIX is rising. This eliminates the wasteful hedging during calm periods that accounts for the bulk of S1's opportunity cost.

Third, the smooth blending approach (S3) offers a middle ground but is less effective than the binary VoV switch, reducing total cost by only 28%. This suggests that partial hedging during low-VoV regimes still incurs substantial opportunity cost without proportionate risk reduction.

3.3 Structural Advantages of the 50/50 Benchmark

A persistent puzzle is that no VT strategy reliably outperforms a simple 50/50 SPY/GLD portfolio on a Sharpe-ratio basis, despite VT's superior drawdown characteristics. We resolve this puzzle by identifying three structural advantages embedded in the 50/50 benchmark.

First, SPY and GLD exhibit near-zero correlation ($\rho = 0.057$) over the extended 2006–2024 sample period,[2] producing a portfolio volatility of 11.47%—30.7% below SPY's standalone volatility of 16.56%. Second, monthly rebalancing of the 50/50 portfolio generates a structural premium of 54 basis points per annum (CAGR of 10.02% vs. 9.49% for buy-and-hold), consistent with the theoretical prediction of Booth and Fama (1992) and confirmed by our weight-sweep analysis across allocations from 30/70 to 70/30 (experiment K846). This premium arises mechanically from systematic buy-low/sell-high behavior when rebalancing to fixed weights. Third, gold provides crisis alpha: during the four VoV regimes identified by VVIX, gold consistently appreciates during equity drawdowns—the "HighVoV_Rising" regime where VT provides its greatest benefit is precisely the regime where gold delivers positive returns, creating a natural hedge that requires no signal, no transaction, and no parameter estimation.

These three structural advantages combine to generate approximately 54–80 bps per annum (the rebalancing premium alone plus the diversification benefit). While S2 achieves a higher Sharpe ratio than 50/50 (0.63 vs. 0.50), this reflects fundamentally different risk exposures: S2 is predominantly equity with state-contingent hedging, whereas 50/50 derives its risk-adjusted return from cross-asset diversification and mechanical rebalancing. Comparing them on Sharpe alone obscures this distinction.

3.4 Regime Analysis

The insurance premium varies dramatically across VoV regimes. During "HighVoV_Falling" periods (7.7% of the sample), S1's opportunity cost spikes to 19.95%—these are the post-

[2] The correlation and rebalancing premium estimates use the 2006–2024 sample (from GLD inception) rather than the 2012–2024 VVIX-reliable period. The 2012–2024 sub-period yields $\rho = 0.04$ and a rebalancing premium of 48 bps, broadly consistent with the full-sample estimates.

crisis recoveries where VT misses the rebound. In contrast, during "LowVoV_Falling" periods (45.5% of the sample), S1's opportunity cost is a more modest 2.54%. The VoV-conditional strategy S2 eliminates this regime asymmetry by construction: it holds 100% equity during low-VoV regimes, incurring zero opportunity cost in the 76% of days when VoV is below the threshold.

3.5 Cross-OOS Robustness

We evaluate robustness using four non-overlapping two-year windows. S2 outperforms BH SPY in only 1 of 4 periods (2019–2020, the COVID crash), while 50/50 SPY/GLD outperforms in 2 of 4. Diebold-Mariano tests[3] confirm that no strategy difference achieves the Harvey et al. (2016) threshold of $|t| > 3.0$ (S1 vs. S0: $t = 2.42$; S2 vs. S0: $t = 0.75$). The limited cross-OOS success rate of 1 in 4 windows suggests that VoV conditioning may be sample-specific. We therefore present S2 as a preliminary finding—a promising direction for future investigation rather than a validated trading rule. That said, the limited success is also consistent with VoV conditioning operating as tail-event insurance that activates infrequently but decisively—analogous to fire insurance that pays off rarely but is not thereby valueless.

Sensitivity analysis across VoV z-score thresholds confirms robustness of the decomposition: at thresholds of 0.5, 1.0, and 1.5, the opportunity cost share remains above 57% in all cases (64%, 57%, and 65%, respectively), and total premium reduction relative to unconditional VT ranges from 62% to 76%. The core finding—that opportunity cost dominates direct cost—is invariant to the specific threshold choice.

4 Discussion

Our decomposition yields three specific implications. First, the conventional view that VT is costly because of high turnover (S1's annualized turnover is 872%) is misleading.

[3] DM tests use QLIKE loss on squared returns as the loss function, with Newey-West HAC standard errors (bandwidth = $\lfloor 4(T/100)^{2/9} \rfloor$). We report two-sided t-statistics and apply the Harvey et al. (2016) multiple-testing threshold of $|t| > 3.0$.

At 5 bps per trade, this turnover generates only 43 bps of annual direct cost—less than a typical ETF expense ratio. The true cost is the 420 bps of foregone equity returns, and this distinction matters for strategy design: reducing trading frequency or using futures addresses at most 9% of the problem.

Second, VVIX conditioning does not improve the volatility forecast itself; rather, it identifies *when forecasting matters*. During low-VoV regimes, the volatility forecast is accurate but irrelevant—the best action is to remain fully invested, consistent with Liu et al. (2019)'s finding that VT destroys value in smooth sailing environments. VoV conditioning effectively converts VT from an always-on insurance policy to a state-contingent one, dramatically reducing the premium. This connects to the broader "volatility-of-volatility risk" literature (Bollerslev et al., 2009; Huang et al., 2019; Todorov, 2010; Bekaert and Hoerova, 2014) but applies it to a portfolio construction context rather than asset pricing.

Third, the rebalancing premium quantification reveals a structural benchmark advantage: a 50/50 SPY/GLD portfolio generates 54 bps of "free" alpha annually through mechanical rebalancing, requiring zero forecasting ability. While S2 achieves a higher Sharpe ratio than 50/50 (0.63 vs. 0.50), this comparison is not apples-to-apples: S2 is 100% equity with occasional VT, while 50/50 includes gold diversification that provides crisis alpha without parameter estimation. On a CAGR basis, 50/50 (7.89%) approaches S1 (7.11%) with far less complexity, and its structural rebalancing premium creates a floor that VT must overcome to deliver net positive alpha.

Several limitations warrant discussion. The VVIX-reliable period begins only in 2012, limiting our sample to 13 years. Cross-OOS analysis uses 4 windows (not the desired 5), and VoV conditioning succeeds in only 1 of 4, raising the possibility that S2's full-sample performance is sample-specific. The VoV conditioning rule ($z > 1.0$ with VIX rising) has not been validated through nested out-of-sample testing; future work should apply walk-forward optimization to confirm robustness. The 5 bps transaction cost assumption is conservative; institutional investors may face lower costs, marginally reducing the direct component. Our analysis focuses on SPY; extension to other asset classes and international markets—particularly those with different volatility dynamics such as emerging markets with amplified VIX sensitivity—is warranted and may reveal different opportunity/direct cost ratios.

5 Conclusion

We decompose the volatility targeting insurance premium into opportunity cost and direct (transaction) cost components. The key finding—that 91% of the cost is opportunity cost, not transaction cost—reframes the VT design problem. The path to cheaper VT lies not in lower-friction execution but in smarter regime identification. VVIX-conditional targeting reduces cost by 74% in the full sample by hedging only when the volatility-of-volatility signals genuine regime uncertainty, though cross-OOS analysis (1 of 4 non-overlapping windows) indicates this benefit may be sample-specific. The structural rebalancing premium embedded in 50/50 SPY/GLD creates a high bar for VT to clear, though VoV-conditional VT achieves a higher Sharpe ratio (0.63 vs. 0.50) via a fundamentally different mechanism (equity timing vs. asset-class diversification). We therefore present VoV-conditional VT as a preliminary but promising direction: the cost decomposition itself is robust across thresholds and periods, while the specific VoV conditioning rule warrants further validation through walk-forward optimization and alternative VoV proxies.

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